



University of Stuttgart
Collaborative
Artificial Intelligence



e l l i s
European Laboratory for Learning and Intelligent Systems



University of Stuttgart
Germany

Machine Perception and Learning for Collaborative Intelligent Systems

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www.collaborative-ai.org 

CNNs in Computer Vision

- CNNs for Object Detection
- CNNs for Semantic Segmentation

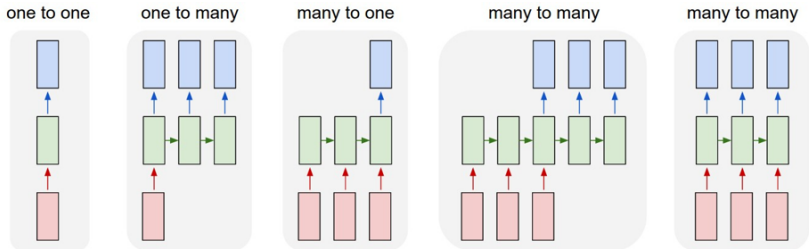
Recurrent Neural Networks

Backpropagation Through Time

An Example Task

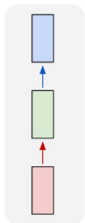
Long Short-Term Memory (LSTM)

Can be used to model a wide range of relationships between (variable- or fixed-length) input and output



Source: Andrej Karpathy

one to one

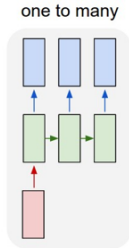


Vanilla Neural Networks

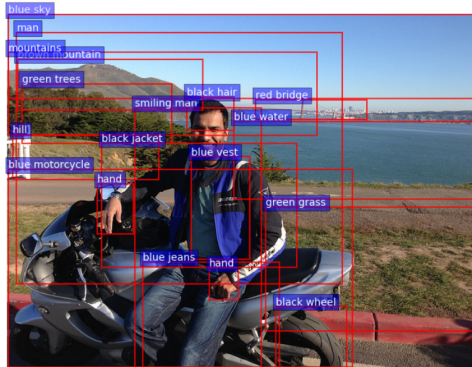


Source: ImageNet [Russakovsky et al., 2015]

“Classical” image classification task

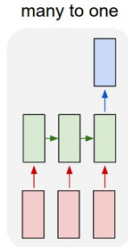


e.g. **Image Captioning:** image $\rightarrow$ sequence of words



Source: Bottom-up and top-down attention for image captioning and visual question answering [Anderson et al., 2018]

Generating text captions for image patches



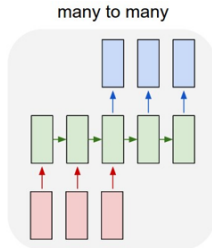
Source: Andre

e.g. **Sentiment Classification:** sequence of words $\rightarrow$ sentiment

e.g. Extracting sentiment from a piece of text

- “Happy New Year. May the Year of the Rat bring you good fortune, cheese in abundance, and immunity to the coronavirus” → **positive**
- “Watching breaking news about the coronavirus 200 infected now! Very sad” → **negative**
- “What are the symptoms of coronavirus and where has it spread?” → **neutral**

Example tweets from the COVIDSenti dataset [Naseem et al., 2021]



©j Karpathy

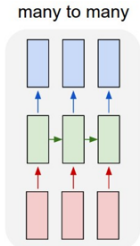
e.g. **Machine Translation:** seq. of words $\rightarrow$ seq. of words

Table 11: Some example translations from PBMT [15], our GNMT system (the "NMT before RL", Table 9), and Human. Source and target sentences (human translations) are from the public benchmark WMT En→Fr (newstest2014) data set. The right-hand column shows the human ratings on a scale of 0 (complete nonsense) to 6 (perfect translation). We disagree with some of the human ratings, e.g., the translation "*Elle a été repéré trois jours plus tard par un promeneur de chien piégé dans la carrière*" contains grammatical mistakes and changes semantics, and is still scored 6. We present it to illustrate the potential problems of the scoring process.

Source	"The reason Boeing are doing this is to cram more seats in to make their plane more competitive with our products," said Kevin Keniston, head of passenger comfort at Europe's Airbus.	
PBMT	"La raison pour laquelle Boeing sont en train de faire, c'est de concentrer davantage de sièges pour prendre leur avion plus compétitive avec nos produits", a déclaré Kevin M. Keniston, chef du confort des passagers de l'Airbus de l'Europe.	3.0
GNMT	"La raison pour laquelle Boeing fait cela est de créer plus de sièges pour rendre son avion plus compétitif avec nos produits", a déclaré Kevin Keniston, chef du confort des passagers chez Airbus.	6.0
Human	"Boeing fait ça pour pouvoir caser plus de sièges et rendre ses avions plus compétitifs par rapports à nos produits", a déclaré Kevin Keniston, directeur de Confort Passager chez l'avionneur européen Airbus.	6.0

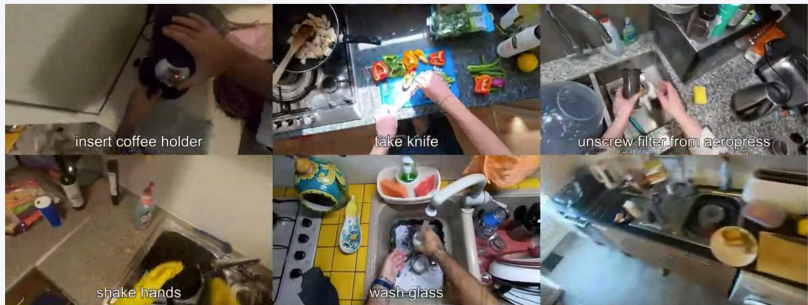
Source: Google's Neural Machine Translation System: Bridging the Gap between Human and Machine Translation [Wu et al., 2016]

Translating text between different languages



e.g. **Video Classification** on frame level

e.g. Frame-level activity recognition



Source: Epic-Kitchens dataset [Damen et al., 2021, 2018]

100 iterations: Model samples random jumbles

```
tyntd-iafhatawiaoihrdemot lytdws e ,tfti, astal f ogoh eoase rrranbyne 'nhthnee e  
plia tkllrgd t o idoe ns,smtt h ne etie h,hregtrs nigtkie,aoaenns lng
```

300 iterations: Model understands quotes and periods

```
"Tmont thithey" fomesscerliund  
Keushey. Thom here  
sheulke, anmerenith ol sivh I lalterthend Bleipile shuwv fil on aseterlome  
coaniogennc Phe lism thond hon at. MeiDimorotion in ther thize."
```

500 iterations: Model can spell short and common words

```
we counter. He stutn co des. His stanted out one ofler that concossions and was  
to gearang reay Jotrets and with fre colt ofp paitt thin wall. Which das stimn
```

Source: Andrej Karpathy

700 iterations: English-like text

Aftair fall unsuch that the hall for Prince Velzonski's that me of
her hearly, and behs to so arwage fiving were to it beloge, pavu say falling misfort
how, and Gogition is so overelical and ofter.

1,200 iterations: Quotations, questions, exclamation marks

"Kite vouch!" he repeated by her
door. "But I would be done and quarts, feeling, then, son is people...."

2,000 iterations: Properly spelled words, quotations, names, etc.

"Why do what that day," replied Natasha, and wishing to himself the fact the
princess, Princess Mary was easier, fed in had oftened him.
Pierre aking his soul came to the packs and drove up his father-in-law women.

Source: Andrej Karpathy

Naturalism and decision for the majority of Arab countries' capitalide was grounded by the Irish language by [[John Clair]], [[An Imperial Japanese Revolt]], associated with Guangzham's sovereignty. His generals were the powerful ruler of the Portugal in the [[Protestant Immineners]], which could be said to be directly in Cantonese Communication, which followed a ceremony and set inspired prison, training. The emperor travelled back to [[Antioch, Perth, October 25|21]] to note, the Kingdom of Costa Rica, unsuccessful fashioned the [[Thrales]], [[Cynth's Dajoard]], known in western [[Scotland]], near Italy to the conquest of India with the conflict. Copyright was the succession of independence in the slop of Syrian influence that was a famous German movement based on a more popular servicious, non-doctrinal and sexual power post. Many governments recognize the military housing of the [[Civil Liberalization and Infantry Resolution 265 National Party in Hungary]], that is sympathetic to be to the [[Punjab Resolution]]
(PJS)[<http://www.humah.yahoo.com/guardian.cfm/7754800786d17551963s89.htm> Official economics Adjoint for the Nazism, Montgomery was swear to advance to the resources for those Socialism's rule, was starting to signing a major tripad of aid exile.]]

Source: Andrej Karpathy

PANDARUS:

Alas, I think he shall be come approached and the day
When little strain would be attain'd into being never fed,
And who is but a chain and subjects of his death,
I should not sleep.

Second Senator:

They are away this miseries, produced upon my soul,
Breaking and strongly should be buried, when I perish
The earth and thoughts of many states.

DUKE VINCENTIO:

Well, your wit is in the care of side and that.

Second Lord:

They would be ruled after this chamber, and
my fair nues begun out of the fact, to be conveyed,
Whose noble souls I'll have the heart of the wars.

Clown:

Come, sir, I will make did behold your worship.

VIOLA:

I'll drink it.

Source: Andrej Karpathy

For $\bigoplus_{i=1,\dots,m}$ where $\mathcal{L}_{m*} = 0$, hence we can find a closed subset $\mathcal{H}$ in $\mathcal{H}$ and any sets $\mathcal{F}$ on X , U is a closed immersion of S , then $U \rightarrow T$ is a separated algebraic space.

Proof. Proof of (1). It also start we get

$$S = \text{Spec}(R) = U \times_X U \times_X U$$

and the comparicoly in the fibre product covering we have to prove the lemma generated by $\coprod Z \times_U U \rightarrow V$. Consider the maps M along the set of points Sch_{fppf} and $U \rightarrow U$ is the fibre category of S in U in Section, ?? and the fact that any U affine, see Morphisms, Lemma ?? . Hence we obtain a scheme S and any open subset $W \subset U$ in $\text{Sh}(G)$ such that $\text{Spec}(R') \rightarrow S$ is smooth or an

$$U = \bigcup U_i \times_{S_i} U_i$$

which has a nonzero morphism we may assume that f_i is of finite presentation over S . We claim that $\mathcal{O}_{X,x}$ is a scheme where $x, x', s'' \in S'$ such that $\mathcal{O}_{X,x'} \rightarrow \mathcal{O}_{X',s''}$ is separated. By Algebra, Lemma ?? we can define a map of complexes $\text{GL}_{S'}(x'/S'')$ and we win. $\square$

To prove study we see that $\mathcal{F}|_U$ is a covering of X' , and T_i is an object of $\mathcal{F}_{X/S}$ for $i > 0$ and $\mathcal{F}_p$ exists and let $\mathcal{F}_i$ be a presheaf of $\mathcal{O}_X$ -modules on $\mathcal{C}$ as a $\mathcal{F}$ -module. In particular $\mathcal{F} = U/\mathcal{F}$ we have to show that

$$\tilde{M}^* = \mathcal{I}^* \otimes_{\text{Spec}(k)} \mathcal{O}_{S,s} - i_X^{-1} \mathcal{F}$$

is a unique morphism of algebraic stacks. Note that

$$\text{Arrows} = (\text{Sch}/S)_{fppf}^{\text{pp}}, (\text{Sch}/S)_{fppf}$$

and

$$V = \Gamma(S, \mathcal{O}) \rightarrow (U, \text{Spec}(A))$$

is an open subset of X . Thus U is affine. This is a continuous map of X is the inverse, the groupoid scheme S .

Proof. See discussion of sheaves of sets. $\square$

The result to prove any open covering follows from the less of Example ?? . It may replace S by $X_{\text{spaces}, \text{étale}}$ which gives an open subspace of X and T equal to S_{Zar} , see Descent, Lemma ?? . Namely, by Lemma ?? we see that R is geometrically regular over S .

Lemma 0.1. Assume (3) and (3) by the construction in the description.

Suppose $X = \lim |X|$ (by the formal open covering X and a single map $\text{Proj}_X(A) = \text{Spec}(B)$ over U compatible with the complex

$$\text{Set}(A) = \Gamma(X, \mathcal{O}_{X, \mathcal{O}_X}).$$

When in this case of to show that $\mathcal{Q} \rightarrow \mathcal{C}_{Z/X}$ is stable under the following result in the second conditions of (1), and (3). This finishes the proof. By Definition ?? (without element is when the closed subschemes are catenary. If T is surjective we may assume that T is connected with residue fields of S . Moreover there exists a closed subspace $Z \subset X$ of X where U in X' is proper (some defining as a closed subset of the uniqueness it suffices to check the fact that the following theorem

(1) f is locally of finite type. Since $S = \text{Spec}(R)$ and $Y = \text{Spec}(R)$.

Proof. This is form all sheaves of sheaves on X . But given a scheme U and a surjective étale morphism $U \rightarrow X$. Let $U \cap U = \coprod_{i=1,\dots,n} U_i$ be the scheme X over S at the schemes $X_i \rightarrow X$ and $U = \lim_i X_i$. $\square$

The following lemma surjective restocomposes of this implies that $\mathcal{F}_{x_0} = \mathcal{F}_{x_0} = \mathcal{F}_{X,\dots,0}$.

Lemma 0.2. Let X be a locally Noetherian scheme over S , $E = \mathcal{F}_{X/S}$. Set $\mathcal{I} = \mathcal{I}_1 \subset \mathcal{I}_n$. Since $\mathcal{I}^n \subset \mathcal{I}^n$ are nonzero over $i_0 \leq p$ is a subset of $\mathcal{I}_{n,0} \circ \mathcal{A}_2$ works.

Lemma 0.3. In Situation ?? . Hence we may assume $q' = 0$.

Proof. We will use the property we see that p is the next functor (??). On the other hand, by Lemma ?? we see that

$$D(\mathcal{O}_X) = \mathcal{O}_X(D)$$

where K is an F -algebra where δ_{n+1} is a scheme over S . $\square$

Source: Andrej Karpathy

Proof. Omitted. □

Lemma 0.1. *Let $\mathcal{C}$ be a set of the construction.*

Let $\mathcal{C}$ be a gerber covering. Let $\mathcal{F}$ be a quasi-coherent sheaves of $\mathcal{O}$ -modules. We have to show that

$$\mathcal{O}_{\mathcal{O}_X} = \mathcal{O}_X(\mathcal{C})$$

Proof. This is an algebraic space with the composition of sheaves $\mathcal{F}$ on $X_{\text{étale}}$ we have

$$\mathcal{O}_X(\mathcal{F}) = \{\text{morph}_1 \times_{\mathcal{O}_X} (\mathcal{G}, \mathcal{F})\}$$

where $\mathcal{G}$ defines an isomorphism $\mathcal{F} \rightarrow \mathcal{F}$ of $\mathcal{O}$ -modules. □

Lemma 0.2. *This is an integer Z is injective.*

Proof. See Spaces, Lemma ?? □

Lemma 0.3. *Let S be a scheme. Let X be a scheme and X is an affine open covering. Let $\mathcal{U} \subset \mathcal{X}$ be a canonical and locally of finite type. Let X be a scheme. Let X be a scheme which is equal to the formal complex.*

The following to the construction of the lemma follows.

Let X be a scheme. Let X be a scheme covering. Let

$$b : X \rightarrow Y' \rightarrow Y \rightarrow Y \rightarrow Y' \times_X Y \rightarrow X.$$

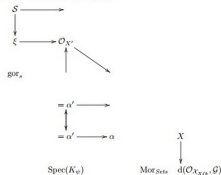
be a morphism of algebraic spaces over S and Y .

Proof. Let X be a nonzero scheme of X . Let X be an algebraic space. Let $\mathcal{F}$ be a quasi-coherent sheaf of $\mathcal{O}_X$ -modules. The following are equivalent

- (1) $\mathcal{F}$ is an algebraic space over S .
- (2) If X is an affine open covering.

Consider a common structure on X and X the functor $\mathcal{O}_X(U)$ which is locally of finite type. □

This since $\mathcal{F} \in \mathcal{F}$ and $x \in \mathcal{G}$ the diagram



is a limit. Then $\mathcal{G}$ is a finite type and assume S is a flat and $\mathcal{F}$ and $\mathcal{G}$ is a finite type f_* . This is of finite type diagrams, and

- the composition of $\mathcal{G}$ is a regular sequence,
- $\mathcal{O}_{X'}$ is a sheaf of rings.

Proof. We have that $X = \text{Spec}(R)$ and $\mathcal{F}$ is a finite type representable by algebraic space. The property $\mathcal{F}$ is a finite morphism of algebraic stacks. Then the cohomology of X is an open neighbourhood of U . □

Proof. This is clear that $\mathcal{G}$ is a finite presentation, see Lemmas ??.

A reduced above we conclude that U is an open covering of $\mathcal{C}$. The functor $\mathcal{F}$ is a field

$$\mathcal{O}_{X, \mathcal{F}} \rightarrow \mathcal{F}_F^{-1}(\mathcal{O}_{X_{\text{étale}}}) \rightarrow \mathcal{O}_{X, \mathcal{F}}^1(\mathcal{O}_{X, \mathcal{F}}^2(\mathcal{O}_{X, \mathcal{F}}^3))$$

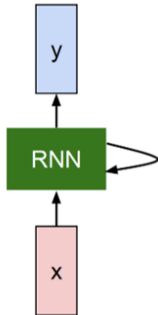
is an isomorphism of covering of $\mathcal{O}_{X, \mathcal{F}}$. If $\mathcal{F}$ is the unique element of $\mathcal{F}$ such that X is an isomorphism.

The property $\mathcal{F}$ is a disjoint union of Proposition ?? and we can filtered set of presentations of a scheme $\mathcal{O}_X$ -algebra with $\mathcal{F}$ are opens of finite type over S . If $\mathcal{F}$ is a scheme theoretic image points. □

If $\mathcal{F}$ is a finite direct sum $\mathcal{O}_{X, \mathcal{F}}$ is a closed immersion, see Lemma ?? . This is a sequence of $\mathcal{F}$ is a similar morphism.

Source: Andrej Karpathy

The internal state consists of a single “hidden” vector h



$$h_t = f_W(h_{t-1}, x_t)$$

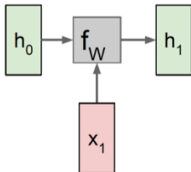
↓

$$h_t = \text{tanh}(W_{hh}h_{t-1} + W_{xh}x_t)$$

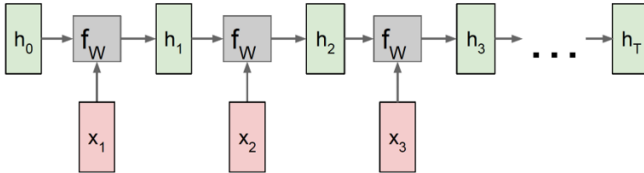
non-linearity (activation function)

$$y_t = W_{hy}h_t$$

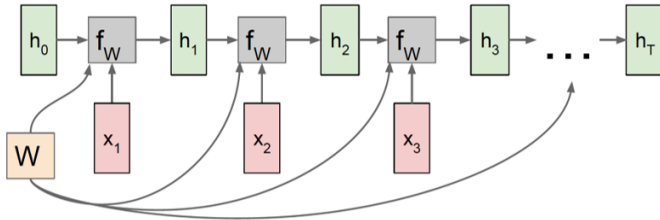
Source: Fei-Fei, Justin Johnson, Serena Yeung



Source: Fei-Fei, Justin Johnson, Serena Yeung

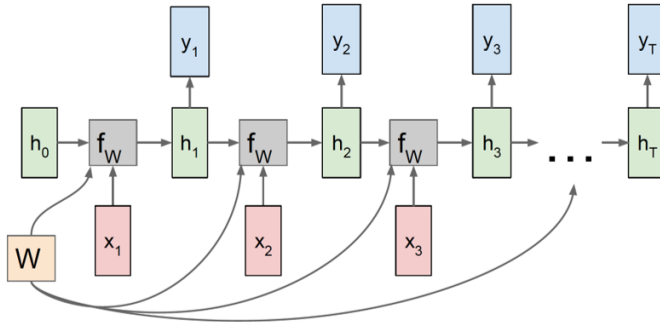


Source: Fei-Fei, Justin Johnson, Serena Yeung



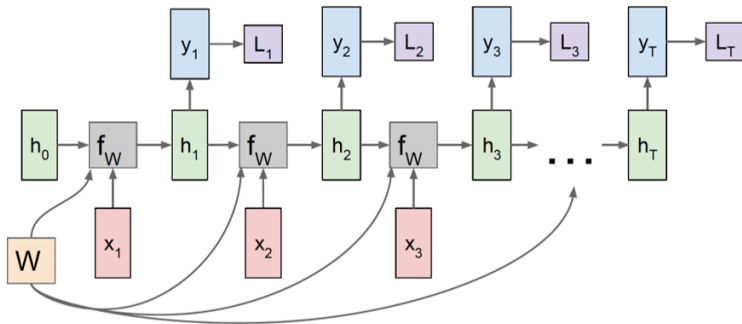
Source: Fei-Fei, Justin Johnson, Serena Yeung

At every time step, the input to f is a unique h_{t-1} and x_t but the **same** weight matrix W



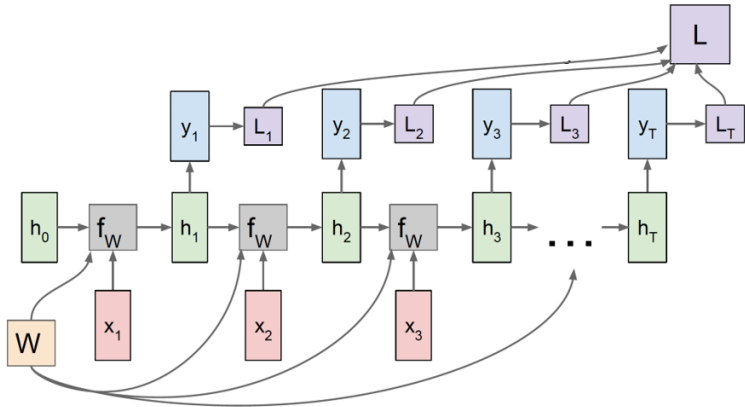
Source: Fei-Fei, Justin Johnson, Serena Yeung

y_t : Outputs computed from h_t at every time step, e.g. a class score



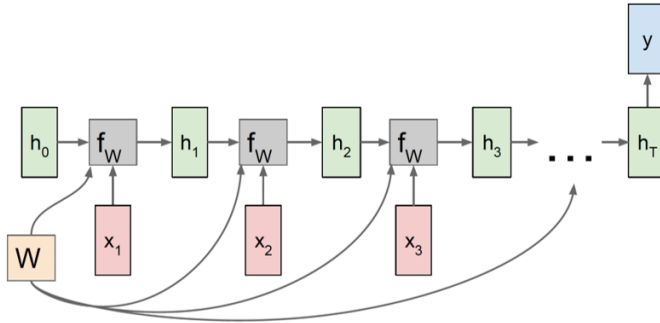
Source: Fei-Fei, Justin Johnson, Serena Yeung

L_t : Intermediate losses calculated from ground truth labels (if available at every time step)



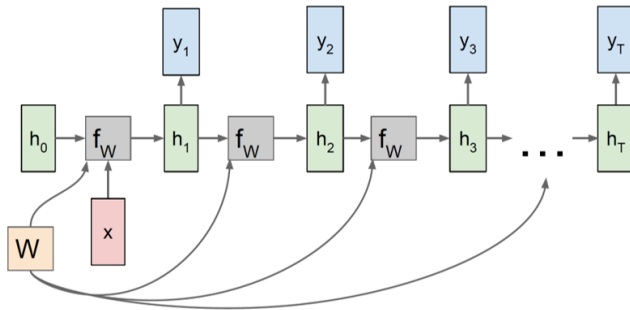
Source: Fei-Fei, Justin Johnson, Serena Yeung

Final loss L is the sum of the intermediate losses



Source: Fei-Fei, Justin Johnson, Serena Yeung

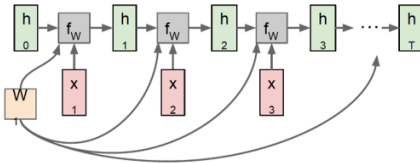
The final hidden state “summarises” all of the context of the entire variable-sized input sequence



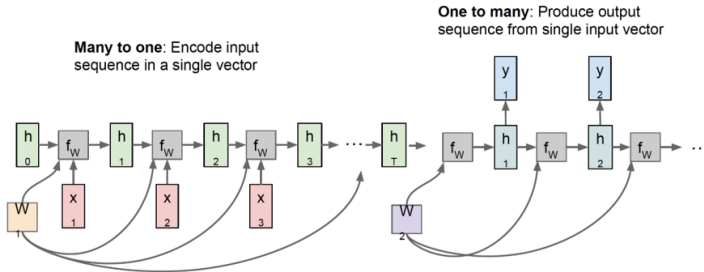
Source: Fei-Fei, Justin Johnson, Serena Yeung

Fixed-sized input is used to initialise the initial hidden state of the model that produces a variable-sized output

Many to one: Encode input sequence in a single vector



Source: Fei-Fei, Justin Johnson, Serena Yeung

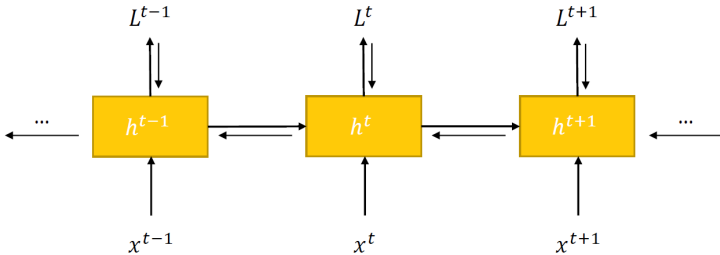


Source: Fei-Fei, Justin Johnson, Serena Yeung

Combination of a many-to-one (encoder) and a one-to-many (decoder) network

Recurrent Neural Networks

Backpropagation Through Time (BPTT)



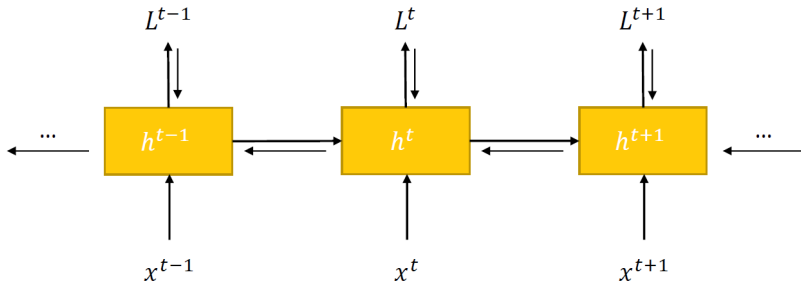
Source: Otmar Hilliges

$$h^t = f(h^{t-1}, x^t, W)$$

$$\hat{y}^t = W_{hy} h^t$$

$$L^t = \|\hat{y}^t - y^t\|^2$$

$$\frac{\partial L}{\partial W} = \sum_{t=1}^S \frac{\partial L^t}{\partial W}$$



Source: Otmar Hilliges

Intuition: Treat unrolled recurrent model as a multi-layer network (with unbounded number of layers) and perform backpropagation

Intuition: Simple linear model, forward pass only:

$$h^t = W^T h^{t-1} \quad (\text{one time step})$$

$$h^t = (W^t)^T h^0 \quad (\text{entire sequence, using power method})$$

If $\exists$ an eigendecomposition of W (if W is symmetric):

$$W = Q\Lambda Q^T$$

then

$$h^t = ((Q\Lambda Q^T)^t)^T h^0 = (Q^T \Lambda^t Q) h^0$$

Λ is the diagonal matrix of the eigenvalues

→ Q : What happens to h^t ?

$$h^t = \sigma(h^{t-1}, x^t, W)$$

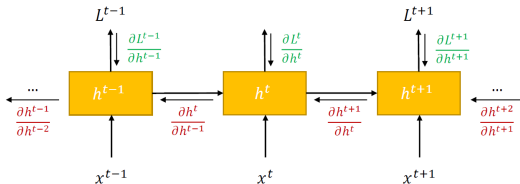
$$\hat{y}^t = W_{hy} h^t$$

$$L^t = \|\hat{y}^t - y^t\|^2$$

$$\frac{\partial L}{\partial W} = \sum_{t=1}^S \frac{\partial L^t}{\partial W}$$

per-timestep loss with
respect to W

$$h^t = \sigma(W_{hh} h^{t-1} + W_{xh} x^t)$$



Source: Otmar Hilliges

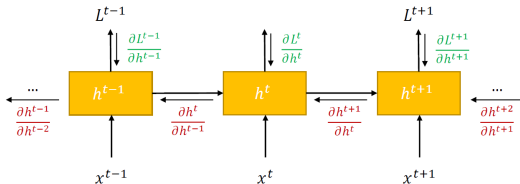
$$h^t = \sigma(h^{t-1}, x^t, W)$$

$$\hat{y}^t = W_{hy} h^t$$

$$L^t = \|\hat{y}^t - y^t\|^2$$

$$\frac{\partial L}{\partial W} = \sum_{t=1}^S \frac{\partial L^t}{\partial W}$$

per-timestep loss with
respect to W



Source: Otmar Hilliges

$$h^t = \sigma(W_{hh}h^{t-1} + W_{hx}x^t)$$

$$\text{Assume: } L^t(f(h^t)) \implies h^t = W_{hh}\sigma(h^{t-1}) + W_{hx}x^t$$

$$\frac{\partial L^t}{\partial W} = \sum_{k=1}^t \frac{\partial L^t}{\partial y^t} \frac{\partial y^t}{\partial h^t} \frac{\partial h^t}{\partial h^k} \frac{\partial^+ h^k}{\partial W}$$

∂^+ : “immediate derivative” $\rightarrow$
we assume that h^{k-1} is a constant
with respect to W
 $\Rightarrow$ value of row i is $\sigma(h^{k-1})$

Temporal components propagate the error through time

$$\frac{\partial L^t}{\partial W} = \sum_{k=1}^t \frac{\partial L^t}{\partial y^t} \frac{\partial y^t}{\partial h^t} \frac{\partial h^t}{\partial h^k} \frac{\partial^+ h^k}{\partial W}$$

∂^+ : “immediate derivative” $\rightarrow$
 we assume that h^{k-1} is a constant
 with respect to W
 $\Rightarrow$ value of row i is $\sigma(h^{k-1})$

Temporal components propagate the error through time

$$\frac{\partial h^t}{\partial h^k} = \prod_{i=k+1}^t \frac{\partial h^i}{\partial h^{i-1}} = \prod_{i=k+1}^t W_{hh}^T \text{diag}(\sigma'(h^{i-1}))$$

(for the specific parametrisation chosen before)

Let λ_1 be the largest singular value of W_{hh}

$$\|diag(\sigma'(h^{i-1}))\| \leq \gamma \in \mathbb{R}$$

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$$\|diag(\sigma'(h^{i-1}))\| \leq \gamma \in \mathbb{R}$$

It is sufficient if $\lambda_1 < \frac{1}{\gamma}$ for gradients to vanish

$$\forall i, \left\| \frac{\partial h^i}{\partial h^{i-1}} \right\| \leq \|W_{hh}^T\| \|diag(\sigma'(h^{i-1}))\| < \frac{1}{\gamma} \gamma < 1$$

Let λ_1 be the largest singular value of W_{hh}

$$\|diag(\sigma'(h^{i-1}))\| \leq \gamma \in \mathbb{R}$$

It is sufficient if $\lambda_1 < \frac{1}{\gamma}$ for gradients to vanish

$$\forall i, \left\| \frac{\partial h^i}{\partial h^{i-1}} \right\| \leq \|W_{hh}^T\| \|diag(\sigma'(h^{i-1}))\| < \frac{1}{\gamma} \gamma < 1$$

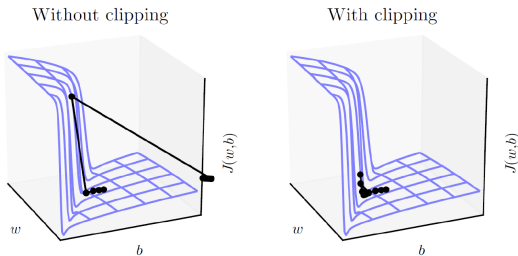
Let $\eta \in \mathbb{R}$ be such that $\forall i, \left\| \frac{\partial h^i}{\partial h^{i-1}} \right\| \leq \eta < 1$ (η exists!)

By induction over i :

$$\left\| \prod_{i=k+1}^t \frac{\partial h^i}{\partial h^{i-1}} \right\| \leq \eta^{(t-k)} \implies \text{if } \lambda_1 < \frac{1}{\gamma}$$

How about exploding gradients?

How about exploding gradients?

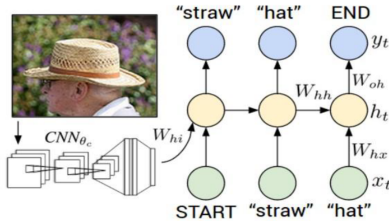


Source: Chapter 10, Deep Learning, Goodfellow 2016

Gradient Clipping: Avoids (drastically) overshooting the local minimum in the energy landscape

Recurrent Neural Networks

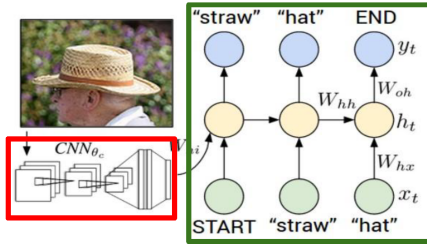
An Example Task



Source: Adapted from Li, Hilliges, Karpathy

- Explain Images with Multimodal Recurrent Neural Networks [Mao et al., 2014]
- Deep Visual-Semantic Alignments for Generating Image Descriptions [Karpathy and Fei-Fei, 2017]
- Show and Tell: A Neural Image Caption Generator [Vinyals et al., 2015]
- Long-term Recurrent Convolutional Neural Networks for Visual Recognition and Description [Donahue et al., 2015]
- Learning a Recurrent Visual Representation for Image Caption Generation [Chen and Zitnick, 2014]

Recurrent Neural Network



Convolutional Neural Network

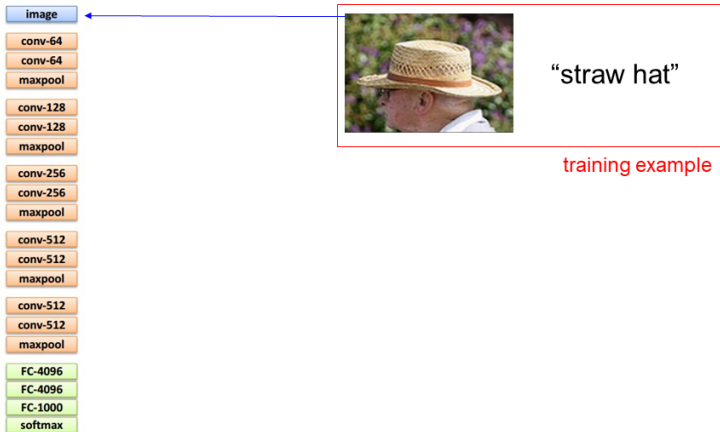
Source: Adapted from Li, Hilliges, Karpathy

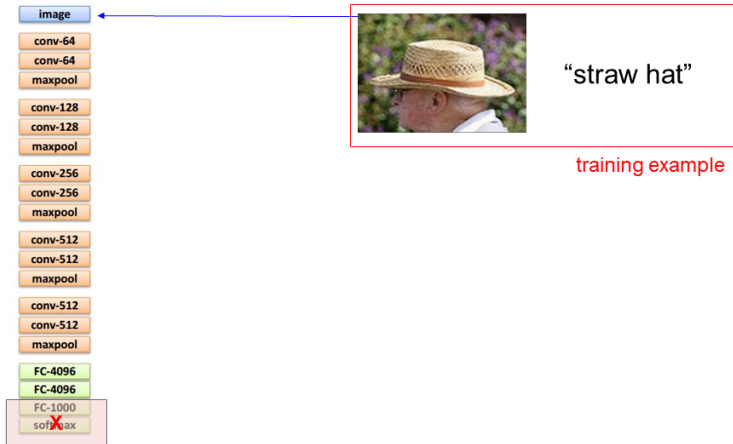
- A convolutional neural network to “parse” the image
- A recurrent neural network for caption generation

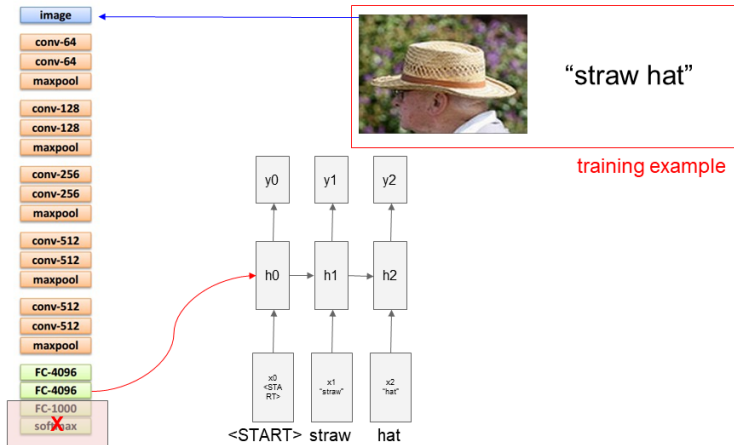


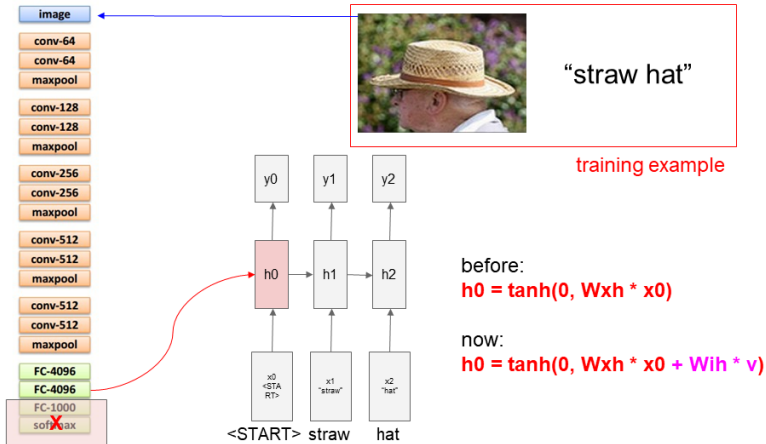
“straw hat”

training example



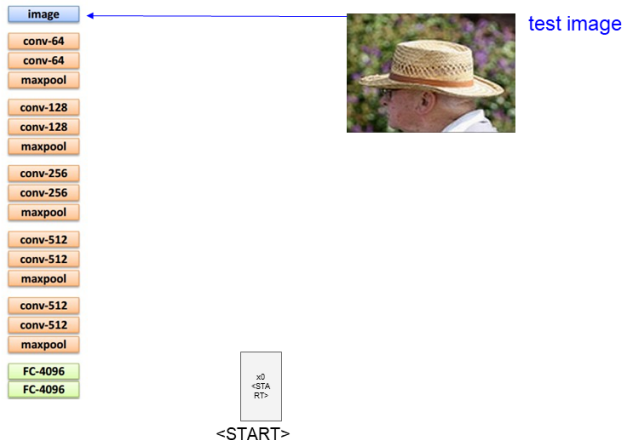


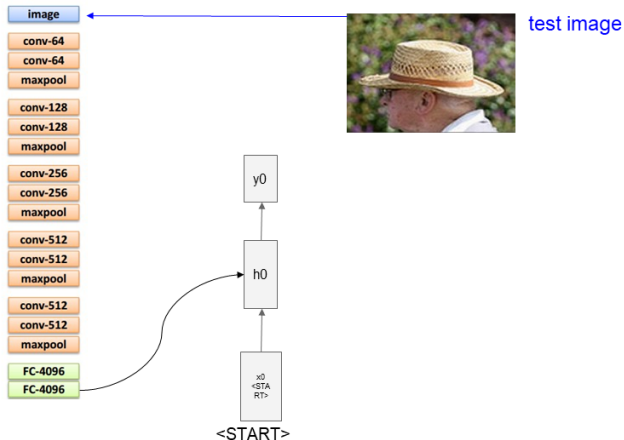


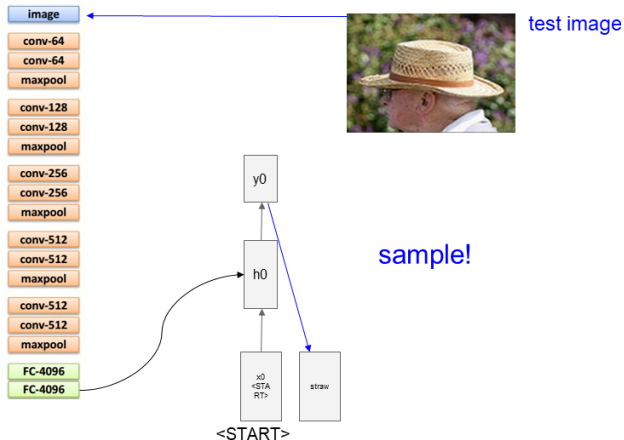


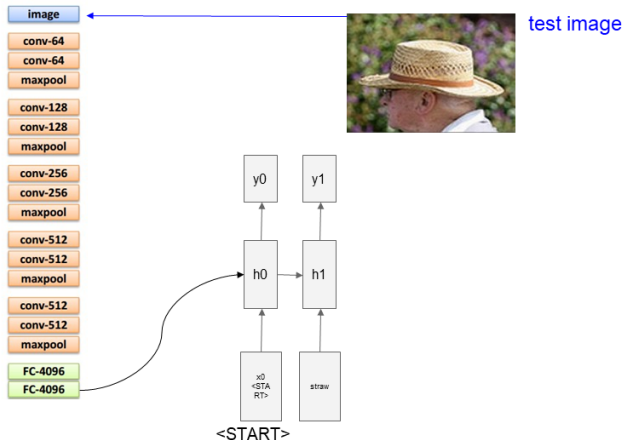


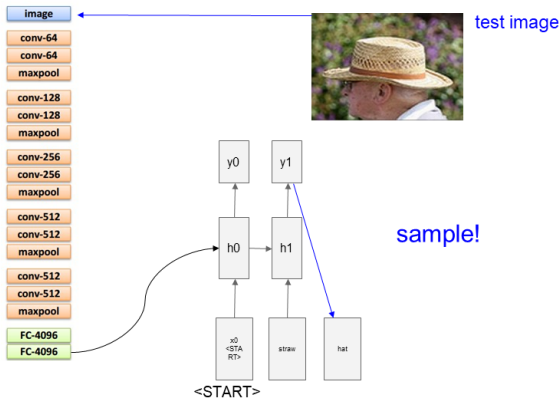
test image



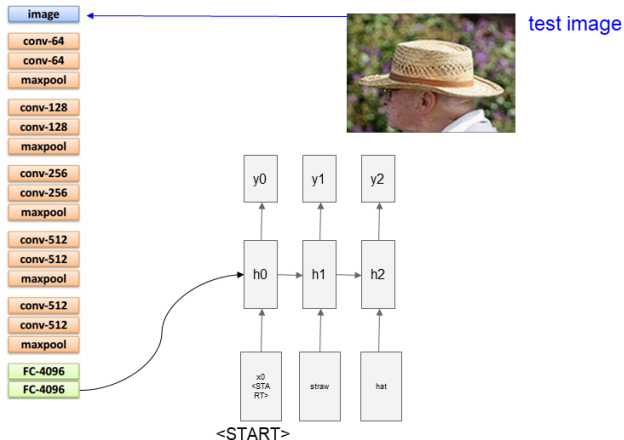


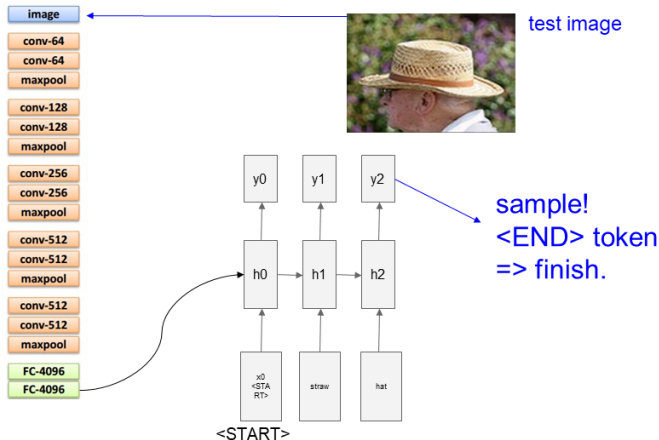






Compute distribution over all words in the vocabulary (English language) and sample from distribution





Wow I can't believe that worked



a group of people standing
around a room with
remotes
logprob: -9.17



a young boy is holding a
baseball bat
logprob: -7.61



a cow is standing in the middle of a street
logprob: -8.84

Wow I can't believe that worked



a cat is sitting on a toilet seat
logprob: -7.79



a display case filled with lots of different types of donuts
logprob: -7.78



a group of people sitting at a table with wine glasses
logprob: -6.71

Slide source: Jeff Donahue

Well, I can kind of see it



a man standing next to a clock on a wall
logprob: -10.08



a young boy is holding a
baseball bat
logprob: -7.65

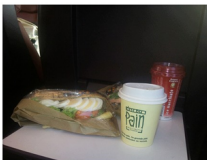


a cat is sitting on a couch with a remote control
logprob: -12.45

Well, I can kind of see it



a baby laying on a bed with a stuffed bear
logprob: -8.66

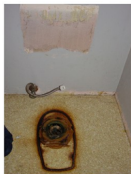


a table with a plate of food and a cup of coffee
logprob: -9.93



a young boy is playing frisbee in the park
logprob: -9.52

Not sure what happened there...



a toilet with a seat up in a
bathroom
logprob: -13.44



a woman holding a teddy bear in front of a mirror
logprob: -9.65



a horse is standing in the middle of a road
logprob: -10.34

Slide source: Jeff Donahue

- Recurrent Neural Networks: <https://towardsdatascience.com/recurrent-neural-networks-rnns-3f06d7653a85>
- The Unreasonable Effectiveness of Recurrent Neural Networks:
<http://karpathy.github.io/2015/05/21/rnn-effectiveness/>
- Minimal character-level language model with a Vanilla Recurrent Neural Network, in Python/numpy
<https://gist.github.com/karpathy/d4dee566867f8291f086>
- The Exploding and Vanishing Gradients Problem in Time Series <https://towardsdatascience.com/the-exploding-and-vanishing-gradients-problem-in-time-series-6b87d5>

Recurrent Neural Networks

Long Short-Term Memory (LSTM)

Question: How to address the vanishing gradient problem and improve gradient flow?

Question: How to address the vanishing gradient problem and improve gradient flow?

Answer: Change RNN architecture

⊙: Hadamard product

Vanilla RNN

$$h_t = \tanh \left(W \begin{pmatrix} h_{t-1} \\ x_t \end{pmatrix} \right)$$

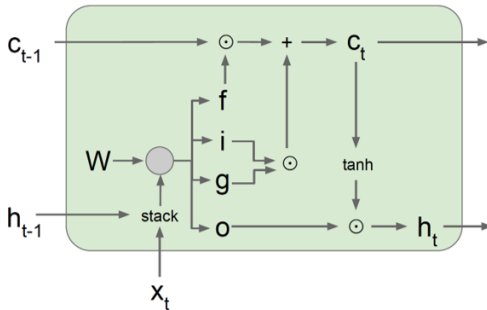
LSTM

$$\begin{pmatrix} i \\ f \\ o \\ g \end{pmatrix} = \begin{pmatrix} \sigma \\ \sigma \\ \sigma \\ \tanh \end{pmatrix} W \begin{pmatrix} h_{t-1} \\ x_t \end{pmatrix}$$

$$c_t = f \odot c_{t-1} + i \odot g$$

$$h_t = o \odot \tanh(c_t)$$

Source: Fei-Fei, Justin Johnson, Serena Yeung



$$\begin{pmatrix} i \\ f \\ o \\ g \end{pmatrix} = \begin{pmatrix} \sigma \\ \sigma \\ \sigma \\ \tanh \end{pmatrix} W \begin{pmatrix} h_{t-1} \\ x_t \end{pmatrix}$$

$$c_t = f \odot c_{t-1} + i \odot g$$

$$h_t = o \odot \tanh(c_t)$$

Source: Fei-Fei, Justin Johnson, Serena Yeung

i: input gate f: forget gate o: output gate g: "gate gate"

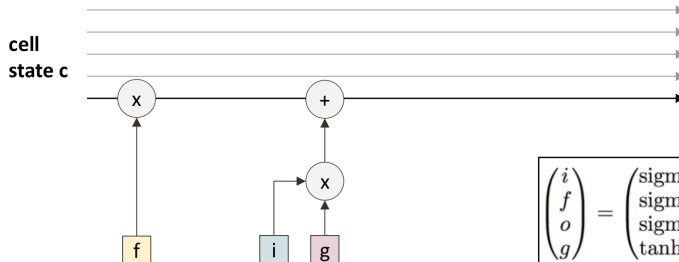


$$\begin{pmatrix} i \\ f \\ o \\ g \end{pmatrix} = \begin{pmatrix} \text{sigm} \\ \text{sigm} \\ \text{sigm} \\ \text{tanh} \end{pmatrix} W^l \begin{pmatrix} h_t^{l-1} \\ h_{t-1}^l \end{pmatrix}$$

$$c_t^l = f \odot c_{t-1}^l + i \odot g$$

$$h_t^l = o \odot \tanh(c_t^l)$$

Source: Fei-Fei, Karpathy

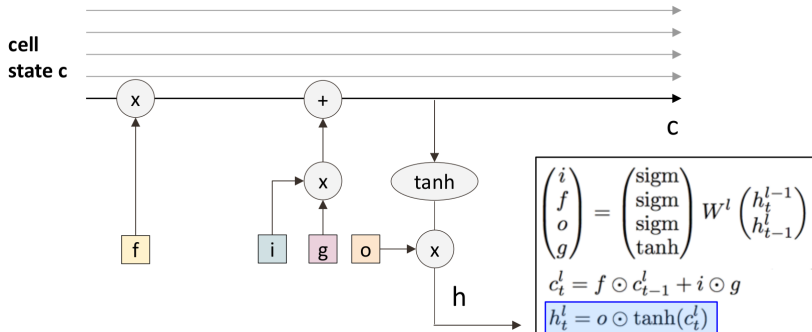


$$\begin{pmatrix} i \\ f \\ o \\ g \end{pmatrix} = \begin{pmatrix} \text{sigm} \\ \text{sigm} \\ \text{sigm} \\ \text{tanh} \end{pmatrix} W^l \begin{pmatrix} h_t^{l-1} \\ h_{t-1}^l \end{pmatrix}$$

$$c_t^l = f \odot c_{t-1}^l + i \odot g$$

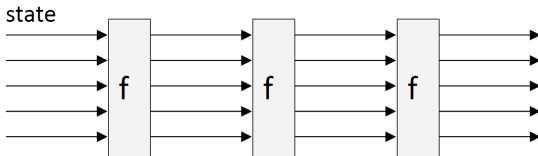
$$h_t^l = o \odot \tanh(c_t^l)$$

Source: Fei-Fei, Karpathy



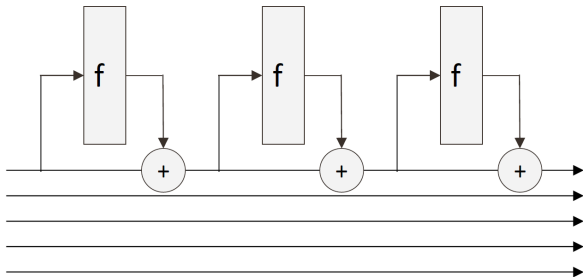
Source: Fei-Fei, Karpathy

RNN

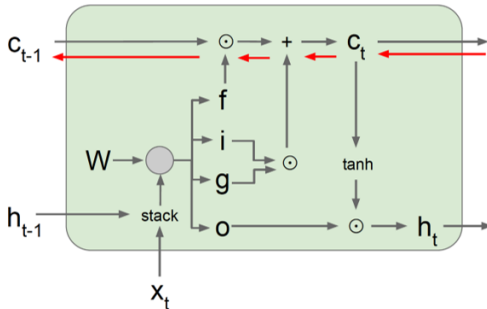


LSTM

(ignoring
forget gates)



Source: Fei-Fei, Karpathy



Backpropagation from c_t to c_{t-1} only elementwise multiplication by f , no matrix multiply by W

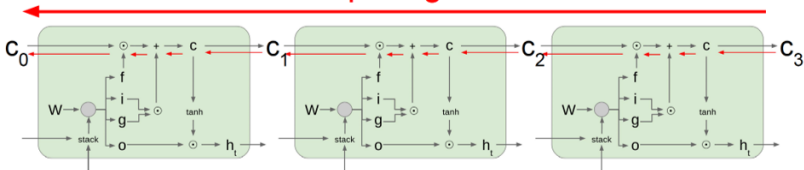
$$\begin{pmatrix} i \\ f \\ o \\ g \end{pmatrix} = \begin{pmatrix} \sigma \\ \sigma \\ \sigma \\ \tanh \end{pmatrix} W \begin{pmatrix} h_{t-1} \\ x_t \end{pmatrix}$$

$$c_t = f \odot c_{t-1} + i \odot g$$

$$h_t = o \odot \tanh(c_t)$$

Source: Fei-Fei, Justin Johnson, Serena Yeung

Uninterrupted gradient flow!

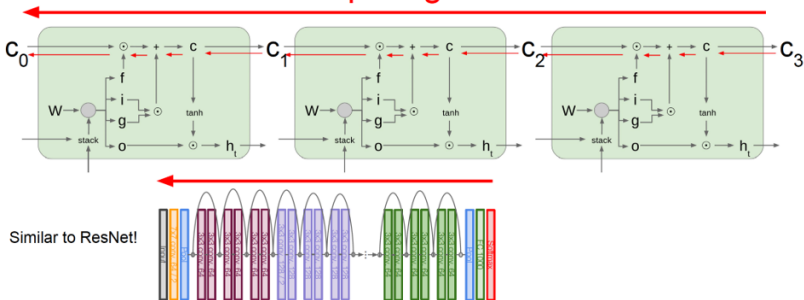


Source: Fei-Fei, Justin Johnson, Serena Yeung

Advantages:

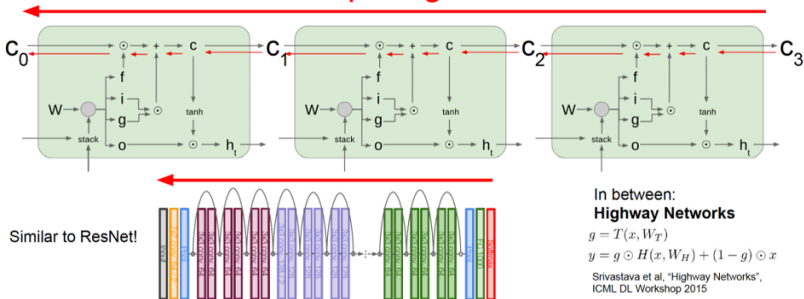
- Element-wise multiplication with value $\in [0, 1]$ is nicer numerically
- Forget gate can vary in each time step $\rightarrow$ upstream cell state gradient is not repeatedly multiplied by the same W
- Gradients don't have to flow through separate $\tanh$ non-linearities at every time step

Uninterrupted gradient flow!



Source: Fei-Fei, Justin Johnson, Serena Yeung

Uninterrupted gradient flow!



Source: Fei-Fei, Justin Johnson, Serena Yeung

Gated Recurrent Units (GRU)

Reset Gate

$$r_t = \sigma(W_r x_t + U_r h_{t-1} + b_r)$$

Update Gate

$$z_t = \sigma(W_z x_t + U_z h_{t-1} + b_z)$$

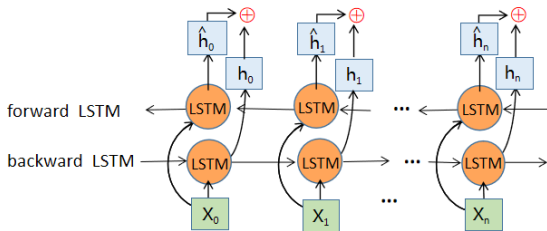
Hidden State

$$\tilde{h}_t = \tanh(W_h x_t + U_h (r_t \odot h_{t-1}) + b_h)$$

$$h_t = z_t \odot h_{t-1} + (1 - z_t) \odot \tilde{h}_t$$

Similar to LSTMs with a forget (reset) gate but have fewer parameters since there is no output gate

BiDirectional LSTM



Source: Ping Huang, Lei Zheng, Ying Wang, Hui Juan Zhu

BiDirectional LSTM

Forward pass:

$$\vec{i}_t = \sigma(\vec{W}_i x_t + \vec{U}_i h_{t-1} + \vec{b}_i)$$

$$\vec{f}_t = \sigma(\vec{W}_f x_t + \vec{U}_f h_{t-1} + \vec{b}_f)$$

$$\vec{o}_t = \sigma(\vec{W}_o x_t + \vec{U}_o h_{t-1} + \vec{b}_o)$$

$$\vec{\tilde{c}}_t = \tanh(\vec{W}_c x_t + \vec{U}_c h_{t-1} + \vec{b}_c)$$

$$\vec{c}_t = \vec{f}_t \odot \vec{c}_{t-1} + \vec{i}_t \odot \vec{\tilde{c}}_t$$

$$\vec{h}_t = \vec{o}_t \odot \tanh(\vec{c}_t)$$

BiDirectional LSTM

Backward pass:

$$\overleftarrow{i}_t = \sigma(\overleftarrow{W}_i x_t + \overleftarrow{U}_i h_{t+1} + \overleftarrow{b}_i)$$

$$\overleftarrow{f}_t = \sigma(\overleftarrow{W}_f x_t + \overleftarrow{U}_f h_{t+1} + \overleftarrow{b}_f)$$

$$\overleftarrow{o}_t = \sigma(\overleftarrow{W}_o x_t + \overleftarrow{U}_o h_{t-1} + \overleftarrow{b}_o)$$

$$\overleftarrow{c}_t = \tanh(\overleftarrow{W}_c x_t + \overleftarrow{U}_c h_{t+1} + \overleftarrow{b}_c)$$

$$\overleftarrow{c}_t = \overleftarrow{f}_t \odot \overleftarrow{c}_{t+1} + \overleftarrow{i}_t \odot \overleftarrow{c}_t$$

$$\overleftarrow{h}_t = \overleftarrow{o}_t \odot \tanh(\overleftarrow{c}_t)$$

Same sequence of data but in reverse order. Two sets of parameters for forward and backward passes

- RNNs allow for flexibility in architecture design (input/output dimensions)
- Vanilla RNNs have simple architecture but do not work very well when the input length is long
- Backward flow of gradients in RNNs can vanish or explode
 - Exploding: Gradient clipping
 - Vanishing: Additive interactions
- (bi-directional) LSTM/GRUs are more powerful in practice: Additive interactions improve gradient flow
- More detailed reading on the different flavours of LSTMs: LSTM: A Search Space Odyssey [Greff et al., 2017]

Transformers

- Embeddings
- Attention Mechanisms
- Training

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- D. Damen, H. Doughty, G. M. Farinella, , A. Furnari, J. Ma, E. Kazakos, D. Moltisanti, J. Munro, T. Perrett, W. Price, and M. Wray. Rescaling egocentric vision: Collection, pipeline and challenges for epic-kitchens-100. *International Journal of Computer Vision (IJCV)*, 2021. URL <https://doi.org/10.1007/s11263-021-01531-2>.
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- S. Hochreiter and J. Schmidhuber. Long short-term memory. *Neural Computation*, 9: 1735–1780, 1997.

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